

Question Set 8

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions

- 1) Copolymerisation as defined in this part of the course, really refers to one type of polymerisation. **RQ**
- Find out which type of polymerisation this is (step-growth or **chain-growth** for example).
 - Explain why this type of copolymerisation (prone to composition drift) is not normally possible for other types of polymerisation. Be as specific in defining types of polymerisation, as possible.
In step-growth, eventually, all the monomer functional groups will polymerise, which means all the monomer will have to be incorporated into polymer in the end (or you have no polymer, according to Carothers!). SO, the concept of composition drift makes no sense in step-growth polymerization.
 - Explain why block-copolymers cannot be synthesised by co-polymerisation
Because block-copolymers mean LONG blocks where there is ONLY one type of monomer. This cannot be achieved if you have two monomers in the system.
 - Given a pair of monomers A and B, is it always possible to synthesise random, alternating, blocky or block copolymers from these monomers?
No, once you fix the monomers and the type of polymerization, this will fix the r (relative reactivity ratios). THESE r values will determine whether you have a random, alternating, etc copolymers.
 - What is the difference between the description "blocky-type" copolymers used in the Lecture notes 4 and the wide-spread definition of block copolymers?
"Blocky" is a word used in this course that has no general acceptance in the literature. We should be talking about "blocky" segments (and use " ", meaning it's a proposed word, not fully accepted word) rather than "blocky" copolymers. "Blocky" segments refers to small segments in a copolymer where there is a large proportion of one monomer and the other one is either absent, or in small fraction. On the other hand, block copolymers (no " ", it's an accepted term) have VERY long segments (and only two, maybe 3, not usually more) where there is ONLY one monomer, exclusively.
 - Are reactivity ratios only defined for free-radical copolymerisation? Once given the ratios, how important is it to know the mechanism? Discuss.
No. Cationic or anionic polymerisation, even ring-opening polymerisation, will have their own r . Once you know the r , you don't need the mechanism to understand what type of copolymer you will get (e.g. alternating, random, etc). BUT understanding the mechanism is key to be able to explain a set of reactivity ratios, r , given a set of monomers.
- 2) The table below shows the reactivity ratios for styrene and other monomers for free radical conditions at 60 °C. **a) answered in red, b) answered in blue. YOU MUST draw the structures as these are essential to answer part b). In an exam you would get the structures.**
- Discuss what type of copolymers will be synthesised starting with a 1:1 ratio of monomers.
 - Explain for the monomer pairs in the table how chemical structure affect copolymerisation reactivity ratios, r .

Free radical copolymerisation reactivity ratios				
M_1	M_2	r_1	r_2	T (°C)
Styrene	Methyl methacrylate	0.52	0.46	60

Styrene	Vinyl acetate	55	0.01	60
Styrene	Maleic anhydride	0.041	0.01	60

Case 1: Styrene, methyl methacrylate:

Both values r_1 and r_2 are essentially the same (very close). This means NO composition drift. But the polymer is NOT random, because the FR from M1 (styrene FR) has 2x the chance of adding onto MMA than onto styrene ($r \sim 0.5$). The same for MMA FR (M2). SO, early, middle and late stages, the same type of polymer will be formed. It's something in between random ($r_1 \sim r_2 \sim 1$) and alternating ($r_1 \sim r_2 \sim 0$). SO a bit of randomness, and a bit of M1M2 and M2M1.

Both FR can stabilise by resonance (for full marks show resonance stabilised FR). If the resonance stabilisation is similar, this will lead to $r_1 \sim r_2$, as is the case here (the system has no preference between M1 FR and M2 FR). However, while styrene has an electron-rich double bond, MMA has an electron poor double bond. If M2 had 2 carbonyl groups attached to the double bond, this would be a VERY electron poor double bond, leading to alternating copolymers $r_1 \sim r_2 \sim 0$. As MMA is not SO electron poor, there is some tendency to alternation and some tendency to randomness ($r_1 \sim r_2 < 1$)

Case 2: styrene , vinyl acetate

$r_1 \gg r_2$: very large composition drift. At the beginning, the monomers have 1:1 ratio. Therefore, r values dominate the kinetics. This means that the polymer formed early is almost a "homopolymerisation" of M1, styrene (not a true homopolymerisation because there will always be some vinyl acetate being incorporated, but this means a "blocky" segments of styrene, separated by 1 unit of vinyl acetate. As the copolymerisation proceeds more and more styrene reacts, but not so much vinyl acetate. Therefore, the concentration of styrene, [styrene] will go down quicker than [vinyl acetate]. Therefore, the "blocky" segments of styrene will get shorter at medium stages of the polymerisation separated by many more 1 units of vinyl acetate. Eventually, as polymerisation proceeds, and [styrene] goes down quite dramatically, we start seeing M2-M2 sequences. At late stages of the polymerisation, all the styrene has reacted, but there is still vinyl acetate left. At this point (very late stages of the polymerisation) there will only be homopolymerisation of vinyl acetate (true homopolymerisation, because there is no styrene left). IMPORTANT: all of these stages lead to different types of copolymers (it is not a long sequence, we are talking about different chains!).

Styrene forms a resonance stabilised FR (show resonance for full marks) while vinyl acetate, does not. This means, the system prefers to proceed through the styrene FR: therefore, when the system is on styrene FR, there is a strong preference to add onto more styrene monomer ($r_1 \gg 1$), while when the system is on vinyl acetate FR, there is a strong tendency to add onto styrene ($r_2 < 1$, $r_2 \sim 0$)

Case 3: styrene and maleic anhydride

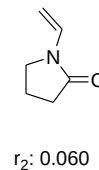
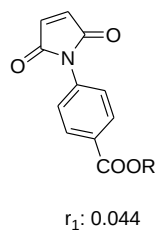
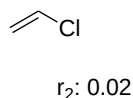
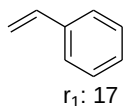
Both values r_1 and r_2 are essentially the same (very close). This means NO composition drift. Because $r_1 \sim r_2 \sim 0$, this means neither FR will tend to add to the same monomer, but cross-polymerise, leading to alternating polymerisation at early, middle and late stages of the polymerisation.

Both FR can stabilise by resonance (for full marks show resonance stabilised FR). If the resonance stabilisation is similar, this will lead to $r_1 \sim r_2$, as is the case here (the system has no preference between M1 FR and M2 FR). However, while styrene has an electron-rich double bond, MMA has a very electron poor double bond (two carbonyl groups attached to the double bond). These systems will form a charge transfer complex (CTC), and it is the CTC that "homopolymerises" leading to an alternating copolymer.

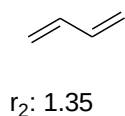
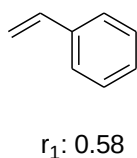
- 3) Explain the following reactivity ratios for free-radical polymerisation based on structural effects. Explain for each of the cases above, what would you observe from a copolymerisation of these monomer pairs starting from a 1:1 feed ratio:

a) Like Case 2 above.

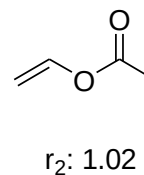
b) Like Case 3 above.



c)



d)



Case c):

$r_1 < r_2$: small composition drift (not as big difference as in case a). At the beginning, the monomers have 1:1 ratio. Therefore, r values dominate the kinetics. However, r_2 is not much larger than 1 and r_1 is not close to 0. This means that the polymer formed early is rather short "blocky" segments of M2 (not a true homopolymerisation because there will always be some M1 being incorporated). As the copolymerisation proceeds more M2 reacts, but less M1. Therefore, the concentration of M2, $[M_2]$ will go down a bit quicker than $[M_1]$. Therefore, the "blocky" segments of M2 will get shorter and we will see more M1-M1 sequences. At very very late stages of the polymerisation, all the M2 has reacted, but there is still M1 left. At this point (very very late stages of the polymerisation) there will only be homopolymerisation of M1 (true homopolymerisation, because there is no M2 left). IMPORTANT: all of these stages lead to different types of copolymers (it is not a long sequence, we are talking about different chains!).

Both FR can stabilise by resonance (for full marks show resonance stabilised FR). If the resonance stabilisation is similar, this will lead to $r_1 \sim r_2$. However, this is not the case. The styrene (M1) FR is a benzylic FR, and the butadiene (M2) FR is an allylic FR. For resonance to take place in M1 FR, aromaticity is disrupted (when the FR resonates within the benzene ring), this incurs in an energy "price", which lowers the stability of the benzylic FR. As this is absent in the allylic FR, the allylic FR is more stable. That's why the system has a slight (but not excessive) tendency to go through M2 FR, which explains $r_2 > 1 > r_1$ (but not $r_2 \gg 1$ and not $r_1 \sim 0$).

Case d)

Both values r_1 and r_2 are essentially the same (very close). This means NO composition drift. Because $r_1 \sim r_2 \sim 1$, the copolymer is random at every stage of the polymerisation.

Neither FR is stabilised by resonance: both FR are equally unstable, therefore there is no tendency, leading to random copolymerisation.